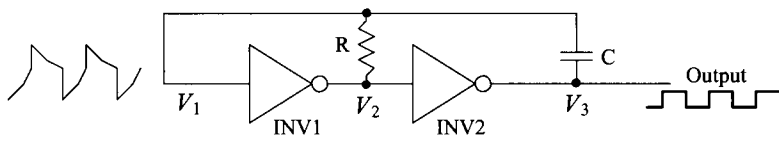


and the frequency of oscillation is

$$f_{osc} = \frac{1}{t_1 + t_2} = \frac{1}{2.2RC} \quad (18.21)$$

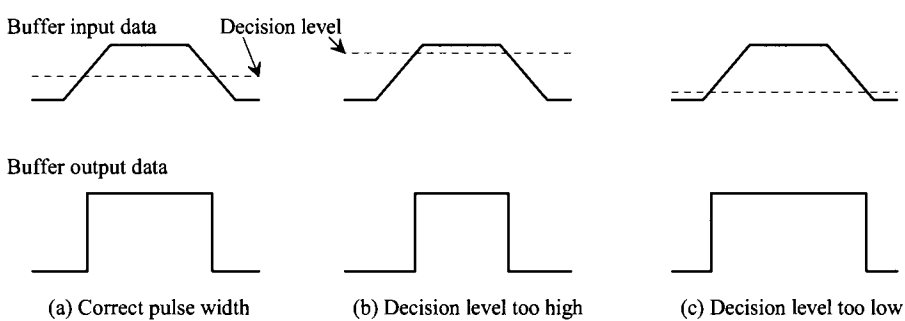
Again, if the resistor and capacitor are bonded out, the ESD diodes on the pads will limit the voltage swing of  $V_1$ .



**Figure 18.11** An astable multivibrator.

### 18.3 Input Buffers

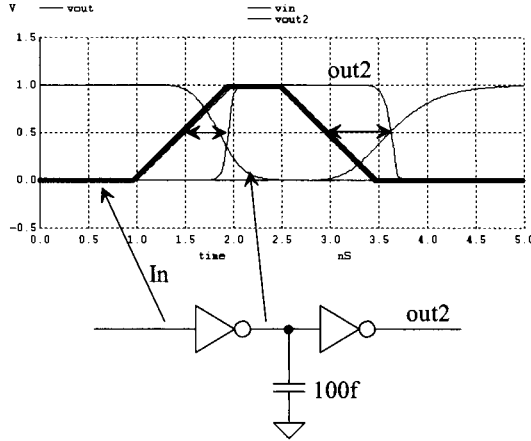
Input buffers are circuits that take a chip's input signal, with imperfections such as slow rise and fall times, and convert it into a clean digital signal for use on-chip. If the buffer doesn't "slice" the data in the correct position, timing errors can occur. For example, consider the waveforms seen in Fig. 18.12. If the input signal is sliced too high or too low, the output signal's width is incorrect. In high-speed systems this reduces the timing budget in the system and can result in errors.



**Figure 18.12** Timing errors in regenerating digital data.

#### 18.3.1 Basic Circuits

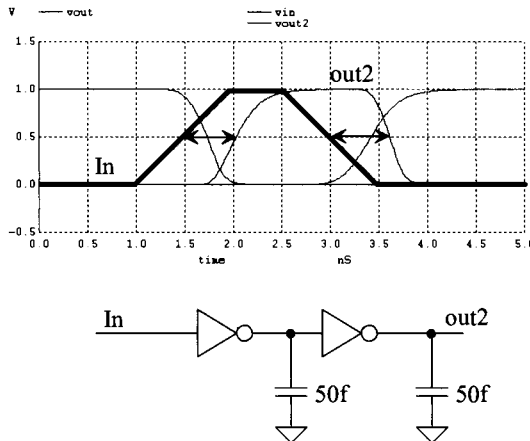
If not careful, simple circuits can introduce unwanted pulse width variations because of differences in rise and fall times. Consider the inverter circuit seen in Fig. 18.13. The switching point voltage of the inverter, because of process,  $V_{DD}$  (voltage), or temperature variations (PVT) will move around. Further, differences in the PMOS and the NMOS resistance will affect the rise and fall times of the inverter's output signal. The output of the first inverter, in Fig. 18.13, sees a much larger load capacitance than the second inverter. The delay time from the input going high to the output going high is roughly



**Figure 18.13** How skew is introduced into a high-speed signal.

400-ps. However, the delay time between the input and the output going low is 600 ps. As the input signal travels through the digital system, the time skew added to the input signal can add up and result in timing errors (the data is received too early or too late at different points in the system). Note that had we matched the loading of each inverter, the propagation delays would be more similar, Fig. 18.14. When the input signal goes high, the output of the first inverter goes low ( $t_{PHL}$ ) and the output of the second inverter goes high ( $t_{PLH}$ ). The total delay is the sum of  $t_{PHL} + t_{PLH}$ . However, when the input signal goes low, the output of the first inverter goes high ( $t_{PLH}$ ) and the output of the second inverter goes low ( $t_{PHL}$ ), resulting in the same sum ( $t_{PHL} + t_{PLH}$ ). If possible, make the number of  $t_{PHL}$  delays in series with a high-speed signal even and equal to the number of  $t_{PLH}$  delays.

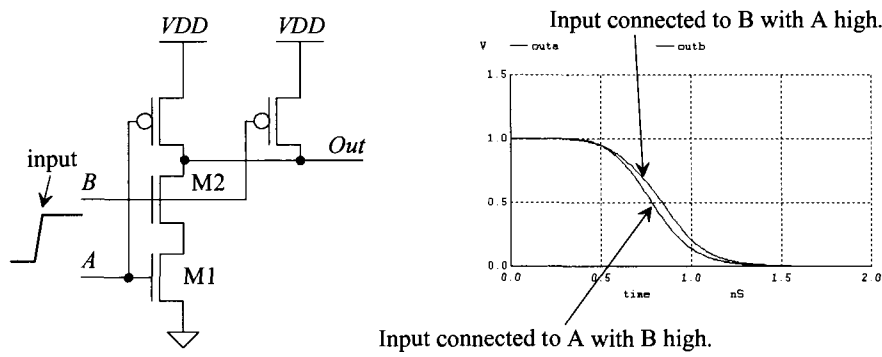
Of course, to better equalize the delays, the second inverter in Fig. 18.14 would need an additional bit of loading (an inverter connected to its output). Also, the input signal should transition faster in an attempt to match the transition times of the inverters.



**Figure 18.14** How equalizing delays can be used to reduce skew.

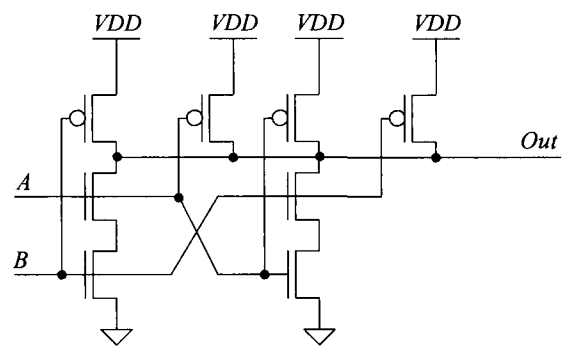
### Skew in Logic Gates

Consider the NAND gate in Fig. 18.15. A different delay will be added to an input signal depending on which input of the NAND gate is used. As seen in the figure, the *A* input, with the *B* input high, propagates to the output slightly quicker than the *B* input (with the *A* input high). For the NAND gate, this difference in propagation delays is only a factor for the NMOS devices (because they are in series with the output of the gate).



**Figure 18.15** The skew introduced by using different inputs.

Figure 18.16 shows how two NAND gates can be used in parallel to eliminate the differences in the propagation delay between inputs. The series NMOS are arranged so that no matter which input is toggled the output changes at the same rate. Note that two of the PMOS devices can be eliminated to simplify the circuit without affecting the propagation delays.



**Figure 18.16** Using two NAND gates in parallel to reduce input-dependent skew.

The previous discussion neglected the effects of the gate's switching point voltage. If the input signals are not transitioning to full logic levels ( $V_{DD}$  and ground) or the rise and fall times are not fast (compared to the gate delay time), then the point where the input circuit switches is critical. This leads us to our next topic.

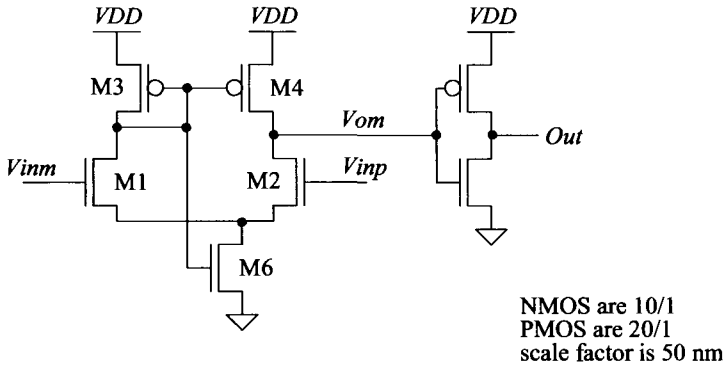
### 18.3.2 Differential Circuits

In order to precisely slice input data, as seen in Fig. 18.12, a reference voltage may be transmitted, on a different signal path, along with the data. Alternatively, the data may be transmitted differentially (an input and its complement). In either case a differential amplifier is needed, Fig. 18.17. A differential amplifier input buffer (which we'll simply call an input buffer from this point on) amplifies the difference between the two inputs. In the simplest case one input to the input buffer is a DC voltage, say 0.5 V ( $V_{inm}$  in Fig. 18.17). When the other input ( $V_{inp}$ ) goes above 0.5 V, the output of the buffer changes states (goes from a low to a high) or

$$V_{inp} > V_{inm} \rightarrow \text{out} = "1"$$
(18.22)

and

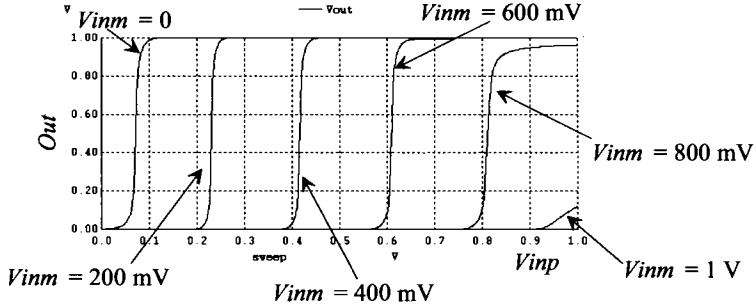
$$V_{inp} < V_{inm} \rightarrow \text{out} = "0"$$
(18.23)



**Figure 18.17** An (n-flavor) input buffer for high-speed digital design.

The diff-amp in Fig. 18.17 is based on the topologies seen in Ch. 22. This circuit is *self-biased* because no external references are used to set the current in the circuit (the gate of M6 is tied up to the gate of M3). When  $V_{inp}$  is larger than  $V_{inm}$ , the current in M2 is larger than the current in M1 ( $V_{GS2} > V_{GS1}$ ). The current in M1 flows through M3 and is mirrored by M4 (and so M4's current is less than M2's current). This causes the diff-amp's output,  $V_{om}$ , to go towards ground (until the current in M2 equals the current in M4) and the output of the inverter,  $Out$ , to go high. Note that the gain from the  $V_{inp}$  input to the output of the circuit is larger than the gain from  $V_{inm}$  to the output (M3, being diode-connected, is a lower resistance than M4). It is generally a good idea to connect the reference voltage to the  $V_{inm}$  input.

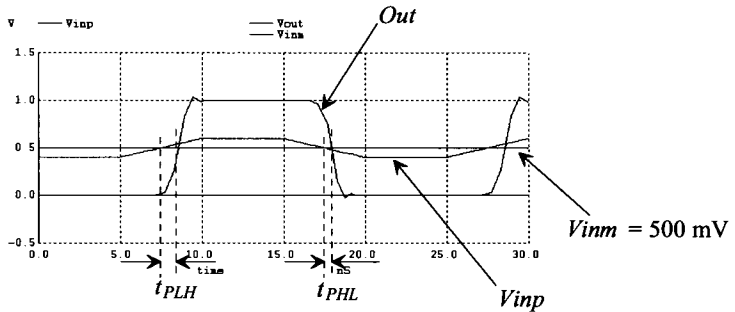
The DC simulation results for the buffer in Fig. 18.17 are shown in Fig. 18.18. The x-axis is the  $V_{inp}$  input swept from 0 to 1 V. The  $V_{inm}$  input is held from 0 to 1 V in 200 mV steps. When  $V_{inm}$  is held at 400 mV and  $V_{inp}$  goes above 400 mV, the output changes from 0 to 1 (although there is a small offset which necessitates that  $V_{inp}$  go to approximately 415 mV before the output transitions).



**Figure 18.18** Simulating the DC behavior of the buffer in Fig. 18.17.

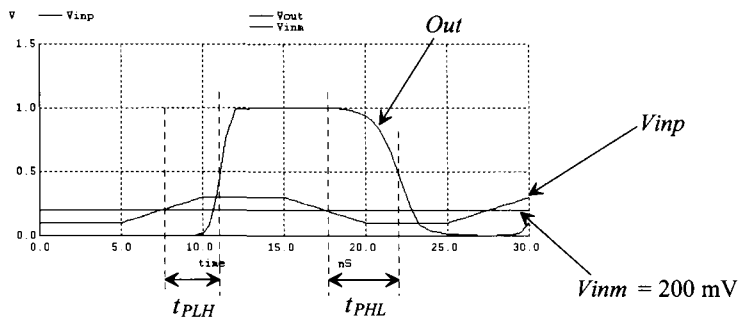
### Transient Response

An example transient response for the buffer in Fig. 18.17 is seen in Fig. 18.19. A very small increase in  $V_{inp}$  above  $V_{inm}$  is required to make the output of the buffer switch states. We have several practical questions that should be answered with variations of this simulation. For example, what are the minimum and maximum values of  $V_{inm}$  allowed. Next, why are the delays between  $V_{inp}$  going high and going low different? Is it because of the offset seen in Fig. 18.18 (the output doesn't precisely switch at a  $V_{inm}$  400 mV)? Let's provide some discussion concerning these concerns.

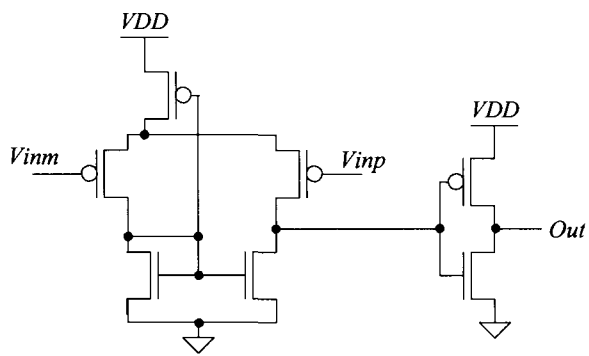


**Figure 18.19** Transient response of the input buffer in Fig. 18.17 driving a 50 fF load.

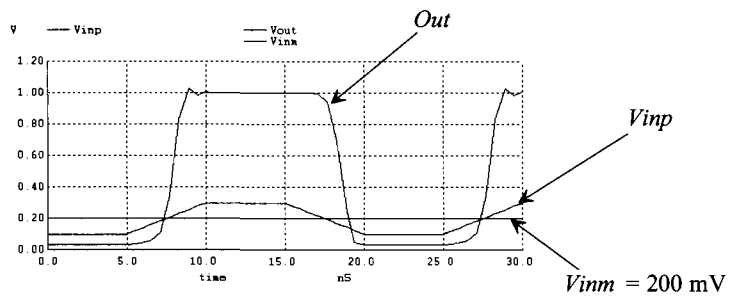
Looking at Fig. 18.17, we can see that if the inputs fall below  $V_{THN}$  ( $= 250$  mV here), then the circuit won't work very quickly (the MOSFETs move into the subthreshold region). So we would expect the propagation delays to increase. Indeed, as seen in Fig. 18.20, the delays go up. Ideally, the delay of the buffer is independent of power supply voltage, temperature, or input signal amplitudes (or pulse shape). To get better performance for lower input level signals, we might use the PMOS version of the buffer in Fig. 18.17, as seen in Fig. 18.21. Resimulating this buffer with the signals seen in Fig. 18.20 gives the results seen in Fig. 18.22. The delays are considerably better, however, there is an offset that appears rather large (because the output changes at the same time as  $V_{inp}$  going past  $V_{inm}$ , indicating that an offset is present). To avoid this



**Figure 18.20** Resimulating with lower input signal voltages.

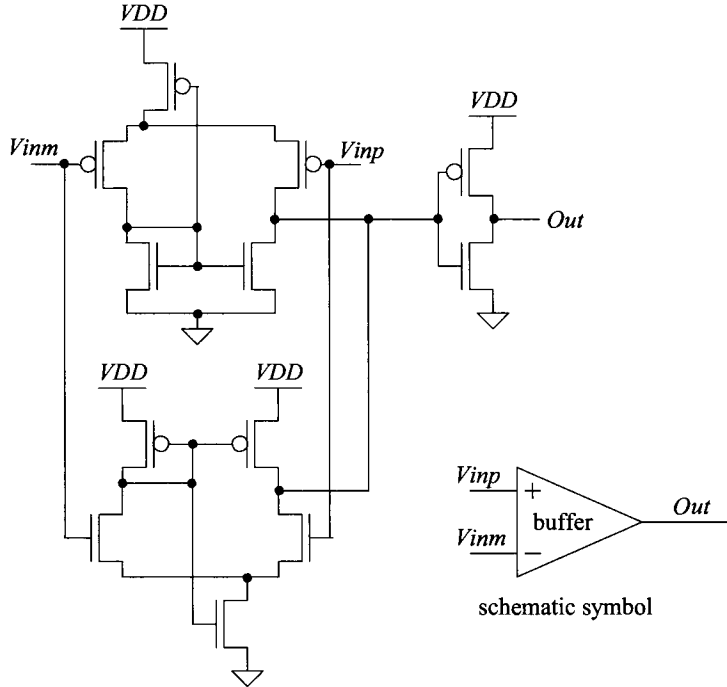


**Figure 18.21** A PMOS input buffer for high-speed digital design.

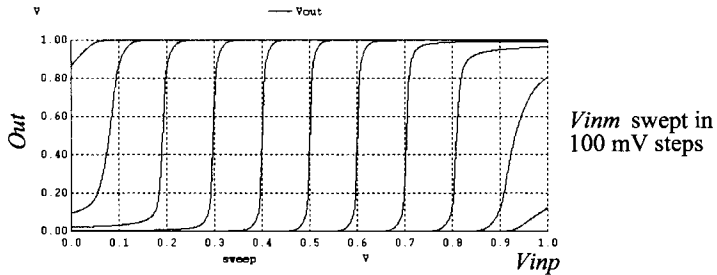


**Figure 18.22** Repeating the simulation in Fig. 18.20 with the buffer in Fig. 18.21.

offset, we might use the NMOS buffer in Fig. 18.17 with the PMOS buffer in Fig. 18.21 to form a buffer that operates well with input signals approaching ground or  $V_{DD}$ . The result is seen in Fig. 18.23. By using the buffers in parallel, the complementary nature results in a buffer that is robust and works over a wide range of operating voltages. Figure 18.24 shows a DC sweep simulation for the buffer with the  $V_{inp}$  input swept and the  $V_{inm}$  changed in increments of 100 mV. Note the smaller offset.



**Figure 18.23** A rail-to-rail input buffer based on the topologies in Figs. 18.17 and 18.21.

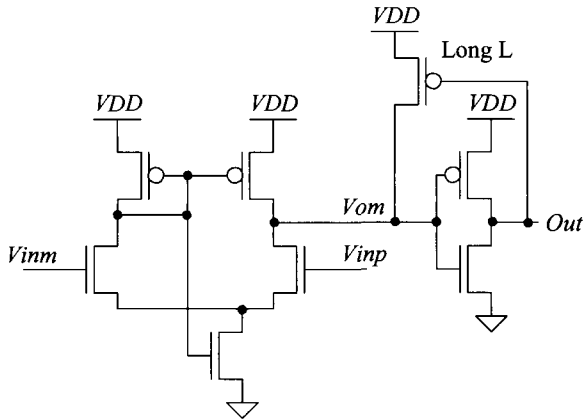


**Figure 18.24** DC characteristics of the buffer in Fig. 18.23.

### Example 18.2

Design a high-speed input buffer based on the topology in Fig. 18.17 with a small amount of hysteresis.

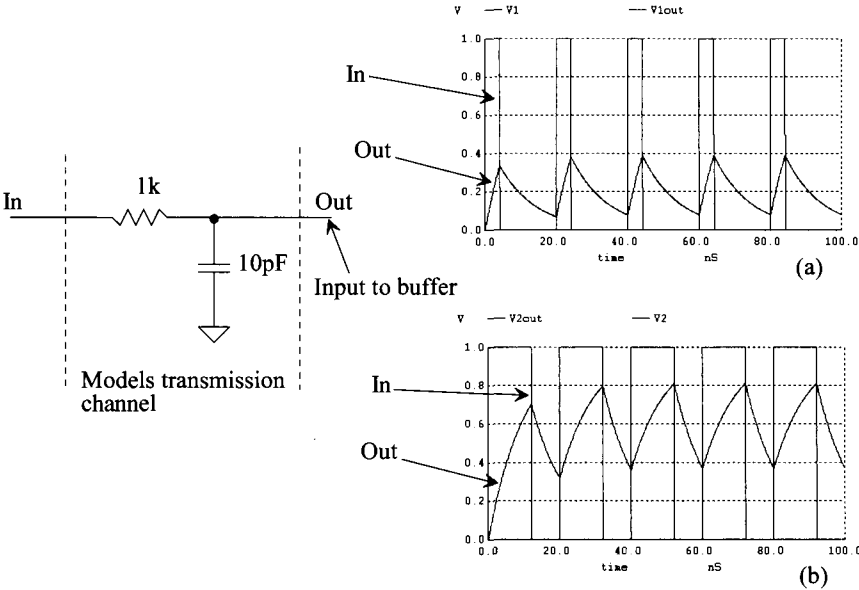
In any circuit that has hysteresis we have some positive feedback. We can introduce positive feedback into the buffer in Fig. 18.17 by adding a long L MOSFET across the output inverter, Fig. 18.25. When the output is low, this turns the long L MOSFET on and make the self-biased diff-amp work harder to pull the input of the inverter to ground. When the output is high, the long L MOSFET is off and doesn't affect the circuit. ■



**Figure 18.25** An input buffer for high-speed digital design with hysteresis.

**18.3.3 DC Reference**

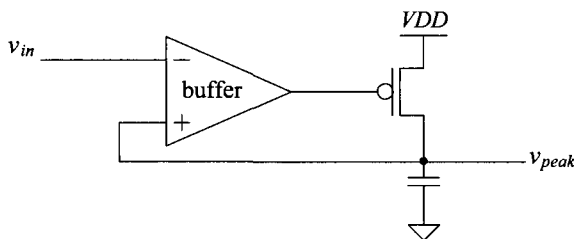
In the previous section we assumed we had a DC voltage,  $V_{inm}$ , at precisely the correct voltage to slice the data (in the middle). In a real system the data varies and the communication channel is bandlimited (behaves like a lowpass filter). The result is that the amplitudes of the data vary depending on the data and the channel frequency response. Consider the simple RC lowpass filter and data used to model a transmission system as seen in Fig. 18.26. The ideal point where we slice the output data changes with the input data. What we need is a circuit that determines the maximum and minimum of an input waveform and outputs the average of the two to slice the buffer's input data in the center.



**Figure 18.26** How the shape of the data changes through a communication channel.

Towards this goal consider the peak detector circuit seen in Fig. 18.27. The buffer from Fig. 18.23 can be used in this circuit. When the input,  $v_{in}$ , goes above the voltage stored on the capacitor,  $v_{peak}$ , the output of the buffer goes low, turning on the long-length MOSFET and pulling the output towards  $VDD$ . As  $v_{peak}$  approaches  $v_{in}$ , the MOSFET starts to shut off. As a result the output voltage across the capacitor,  $v_{peak}$ , corresponds to the peak voltage of the input signal. This peak detector can be used to generate a reference voltage that falls within the middle of the input data, Fig. 18.28. The peak and valley detectors are used to find the minimum and maximum of the input signal. The two resistors average the minimum voltages and feed the result (the DC average of the input) to the bottom buffer circuit. The resistors also are used to leak charge off of the capacitors so that the averaging circuit can follow changes in the input data (the averaging is actually a *running average*). Figure 18.29 shows some simulation results. In the top plots the output signal from Fig. 18.26a is applied to the input of the DC generation circuit. As expected, the output of the DC generation circuit, after a start-up delay, goes right to the middle of the input data.

The speed (response time) of this generation circuit can be increased by reducing the resistors and capacitors in the circuit. However, if the values are reduced too much, a long string of zeroes or ones will cause the DC generation circuit's output (labeled "average" in Fig. 18.28) to go to ground or  $VDD$ . The bottom traces in Fig. 18.29 show the results of applying the output signal from Fig. 18.26b to the DC generation circuit. Again, the output moves quickly to the center of the data. What's more interesting in these figures is a comparison between the original input data and the regenerated output data (labeled *In* and *Out* in Fig. 18.29). The output data pulse widths are considerably different from the input pulse widths (remembering that there is a delay through both the channel RC seen in Fig. 18.26 and the buffer in Fig. 18.28). Ideally, the pulse widths (the data) of the input and output are the same. Looking at the responses in Fig. 18.26a, we see that the effect of the finite channel bandwidth is to distort the channel's output data. *The only solution to this problem is to make the channel transmission bandwidth effectively wider.* An equalizer (discussed in the next chapter) can be used for this purpose at the cost of reduced signal swing. We can also reduce the input impedance of the input buffer to lower the time constant associated with the channel. Let's discuss this second approach.



**Figure 18.27** CMOS peak detector.

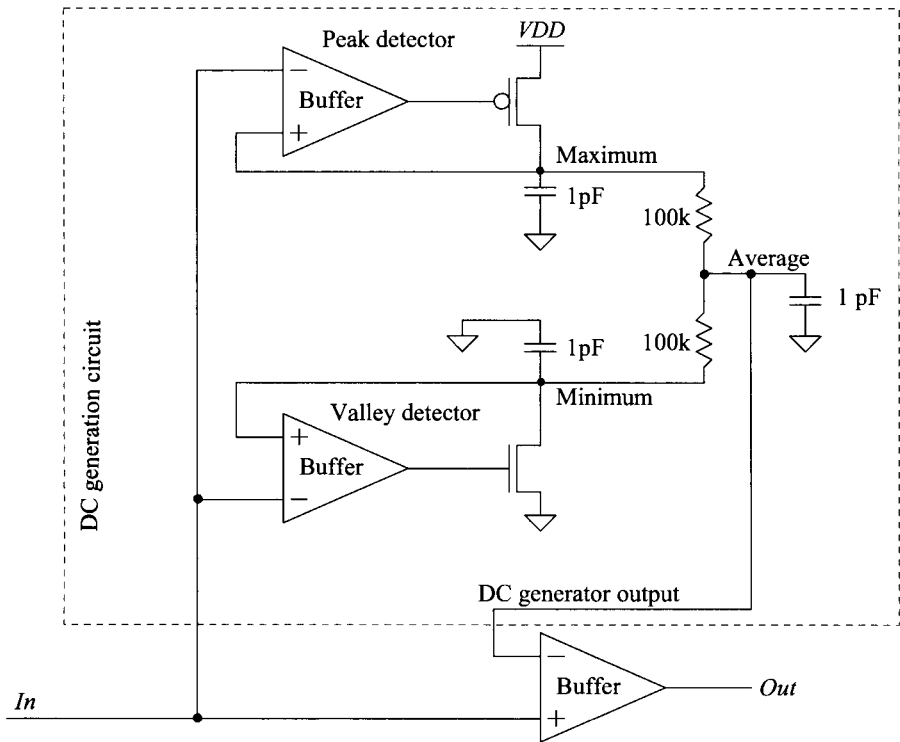


Figure 18.28 A DC generation circuit.

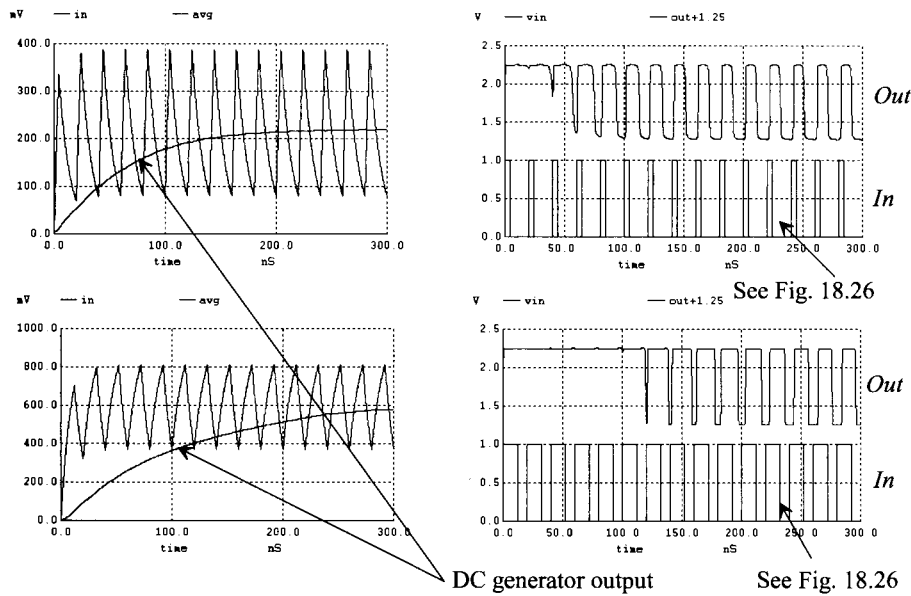
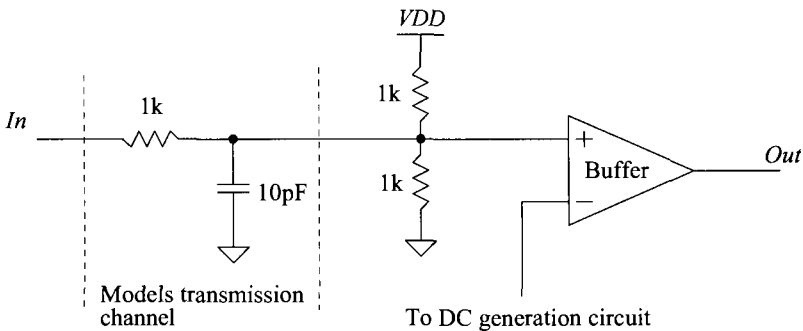


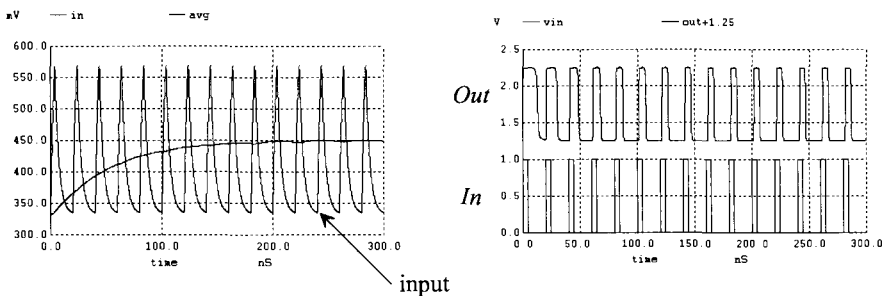
Figure 18.29 The output of the DC restore circuit in Fig. 18.28.

### 18.3.4 Reducing Buffer Input Resistance

A simple solution to increasing the effective bandwidth of the transmission channel is to reduce the input resistance of the input buffer. The time constant associated with the channel in Fig. 18.30 is 10 ns. When we connect the buffer to the transmission channel, the time constant drops to 3.33 ns (the three 1k resistors in parallel multiplied by the 10 pF capacitance). The drawbacks of this approach are increased power dissipation and reduced input signal amplitudes. Figure 18.31 shows how the buffer in Fig. 18.30 behaves with the input signals seen in Fig. 18.26a. The distortion (difference in the pulse widths of ones and zeroes when comparing the communication channel input, *In*, and the buffer output, *Out*) is much better.



**Figure 18.30** Reducing the input resistance of the input buffer.



**Figure 18.31** How reducing the input resistance of the buffer reduces the input signal amplitude (bad) and distortion (good).

Figure 18.32 shows an alternative low input resistance input buffer. When the input goes high, M1 turns on and M4 shuts off. This causes M2 to turn on and M5 to shut off. The output then goes high. When the input goes low, M1 turns off and M4 turns on (with M2 shutting off and M5 turning on). The result: the output goes low. Again, the power burned by M1 and M4 can be high if the transistors aren't sized properly. Further, the input swing is limited to a threshold voltage away from the supply voltages. This reduces the input voltage swing and further enhances the speed.